Technical and instrumentation factors play an important role in obtaining reliable information during electrodiagnostic studies. With contemporary electrodiagnostic equipment, neurophysiologic potentials are detected using a variety of electrodes and undergo differential amplification, filtering, conversion to digital form, and finally, analysis and display. Understanding the signal processing principles, limitations, and sources of errors that can occur during this multistep process can improve the technical quality of studies, minimize preventable errors, and improve clinical interpretation. Part I of this minimonograph reviews the basic principles of action potential generation and overviews electrodiagnostic instrumentation. The concept of waveform frequency content is related to the role of filters in suppressing noise while preserving waveform latency, amplitude, and morphology. The electrical characteristics of various surface and needle electrodes influence instrument design and the nature of the potentials recorded. This is especially important in understanding the differences in motor unit characteristics obtained from monopolar and concentric needle electrodes.
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AAEM MINIMONOGRAPH #16: INSTRUMENTATION AND MEASUREMENT IN ELECTRODIAGNOSTIC MEDICINE—PART I

ANDREW J. GITTER, MD, and WALTER C. STOLOV, MD

Electrodiagnostic medicine involves the recording, display, measurement, and interpretation of action potentials arising from the central nervous system, peripheral nerve, and muscle. Disease and injury alter neuromuscular architecture and physiology, leading to changes in the time-course (onset and duration), amplitude, and configuration of these potentials. These changes are used for diagnosis, to monitor disease progression, and to assess therapeutic interventions. Technical limitations and instrumentation factors can distort neurophysiologic waveforms during their acquisition, resulting in changes that are essentially identical to those seen with disease or injury. The electrodiagnostic physician needs to be aware of these sources of error and understand techniques that can be used to minimize distortion. Part I of this monograph reviews fundamental electrical principles of action potential generation, the concept of the frequency content of waveforms, analog and digital instrumentation basics, and the influence of filters and electrodes on recorded waveforms. Part II continues the discussion of instrumentation issues and covers amplifier basics, waveform display, noise, and electrical stimulation of nerve. A review of patient safety issues related to nerve stimulation and leakage current follows in the appendices to Part II.

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BASIC BIOPHYSICS

Normal nerve and muscle cell membranes have a stable resting transmembrane potential as a result of a uniform distribution of positively charged ions on the outside of the cell membrane and negatively charged ions on the inside. While these charge distributions create a potential difference (voltage) across the membrane, the net potential measured between two points outside of the cell is zero be-

cause the electrical field contributions from the positive and negative charge layers cancel each other. During depolarization of a nerve or muscle cell, current flows along the electrical potential gradients across the membrane, along the interior of the cell, and finally through the extracellular fluid surrounding the nerve or muscle fiber. With depolarization, positive and negative charge distributions along the outside and inside of the membrane are no longer uniformly equal, resulting in the appearance of potential differences between two points outside of the cell. This potential difference varies in time as the membrane undergoes a depolarization/repolarization cycle, creating the extracellularly recorded action potential.

Action potential currents primarily result from the movement of Na^+ and K^+ ions. The tissue between any two points will resist or impede (hence "impedance") the movement of current. The driving force or voltage potential (V), the current (I) produced by the driving force, and the impedance (Z) are related by the equation:

$$\text{Voltage } (V) = \text{Current } (I) \times \text{Impedance } (Z) \quad (1)$$

In electrodiagnostic applications, it is particularly important to understand eq. (1). The potential (voltage) is actually a magnitude that represents a difference of voltage between two points, rather than an absolute value. Thus, current flow will be identical in a particular medium for a difference of 10mV whether that difference exists between two points whose absolute magnitudes are 100 and 90, or 2000 and 1990, or +5 and −5mV. The value of the voltage is positive or negative relative to the ground lead which is taken as the zero reference. In electrodiagnostic studies, action potential magnitudes are expressed in millivolts (mV, 10^{-3} volts) and microvolts (μV, 10^{-6} volts), currents in milliamperes (mA, 10^{-3} amperes), and impedances in kilo-ohms (kΩ, 10^3 ohms) or mega-ohms (MΩ, 10^6 ohms). Time measurements are of the order of milliseconds (ms, 10^{-3} seconds) and microseconds (μs, 10^{-6} seconds). Although the term Hertz (Hz) is in common usage, cycles per second (cps) (note: 1 cps = 1 Hz) will be used throughout the monograph for frequency measurements, because it is becoming the preferred term.

The action potentials which are ultimately measured during clinical studies result from current flowing through tissues in the vicinity of the recording electrodes. The magnitude and polarity of the recorded potential depends on the charge density near the membrane, the distance from the site of depolarization, the properties of the medium, and the location at which the potential is measured relative to depolarized and normally polarized segments of the membrane. The resultant current flows are complex and are influenced by the volume conductor characteristics of soft tissues. Volume conductor effects are complicated and have recently been reviewed by Dumitru and Delisa.[8]

FREQUENCY CHARACTERISTICS OF NEUROPHYSIOLOGIC WAVEFORMS

During clinical studies, neurophysiologic action potentials originating from nerve and muscle are displayed and interpreted as waveforms in which the amplitude varies with time. Some segments of these waveforms change rapidly with time while others change more slowly. The rapidly changing components indicate that the waveform contains high-frequency elements while the slowly changing components result from low-frequency signal elements. The frequency relationships present in a neurophysiologic waveform (or any waveform in general) can be understood by representing the signal as the summation of a series of sine waves of varying frequency, amplitude, and phase. Characterizing a waveform by the various frequency components it contains is known as frequency or Fourier analysis (Fig. 1).

The frequency content of a waveform refers to the number of terms (frequencies) needed to accurately depict the waveform. Frequency analysis of the compound motor action potential (CMAP) and the sensory nerve action potential (SNAP) are shown in Figure 2. The CMAP signal contains primarily low-frequency components below 1000 cps, while the SNAP has a broader range of frequency components which are shifted toward higher frequencies. The different peak frequencies reflect the differences in the fundamental frequency of the CMAP and SNAP, which can be inferred from their respective waveform durations. For the CMAP this occurs between 60 and 100 cps, which is consistent with the long duration (12–15 ms) of the CMAP; for the SNAP the peak is approximately 300 cps, which is consistent with the shorter duration (2–3 ms) of the sensory potential.

To prevent distortion of a neurophysiologic signal, the recording system, including the electrode, must be able to respond to all of the frequency components within the potential and suppress those frequencies that are unrelated to the potential. Action potentials with rapidly changing, high-frequency components include fibrillation potentials, polyphasic motor unit action potentials

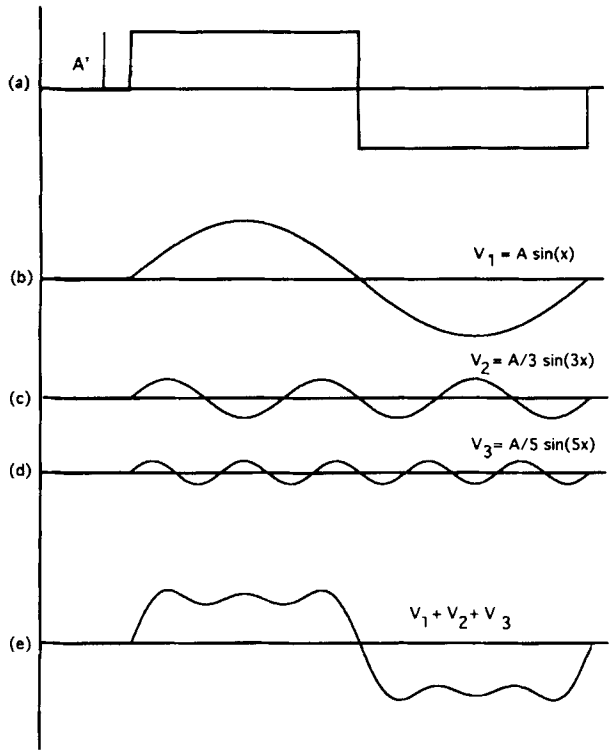


FIGURE 1. An example of the principle of Fourier analysis. Any waveform, the square wave in this example (a) can be decomposed into a series of sine waves of varying frequency, phase, and amplitude. In this example, the first three elements of the series (b, c, d) and their sum (e) are displayed. The addition of more frequency terms would result in a better approximation of the original waveform. ($x = 2\pi ft$ where f is frequency and t time, $A = 4A'/\pi$)

(MUAPs) and the potentials produced by single muscle fibers (single fiber EMG). Potentials with slowly changing components include the CMAP and the SNAP, especially close to the peaks of the positive and negative deflections and during the slow terminal return to baseline, the positive component of positive sharp waves, and somatosensory evoked potentials (SEPs).

BASIC APPARATUS AND INSTRUMENTATION CONCEPTS

During electrodiagnostic testing, the recording system must accurately reproduce the neurophysiologic potentials generated by muscle and nerve fibers. Two electrodes are introduced into the electric field produced by the depolarizing nerve or muscle and the potential is measured as the difference between the two electrode locations. By convention, the electrodes have been named the "active," or G_1, and "reference," or G_2, electrodes. The potentials measured are not those that actually occur, but those that exist as distorted by the presence of the electrode. Electrode size is generally governed by the application and chosen to be both practical and to limit the distortion to an acceptable level.

The basic components of a typical instrument used in EMG, nerve conduction velocity (NCV), and evoked potential studies are illustrated in Figure 3. Neurophysiologic signals range in amplitude from less than 1μV for SEPs to many microvolts for SNAPs to millivolts for MUAPs and CMAPs. These signals exist within an environment filled with electrical noise. Electrical noise from sources such as radio, 60 cps power lines, and other nearby electrical equipment pervade the environment. Even under the best conditions, the noise is typically several orders of magnitude larger than the waveform under study. Successful recording depends on the ability to selectively amplify the signal of interest approximately 50–250,000 times while rejecting the unwanted extraneous noise. After amplification the signals are generally 1–10 V in magnitude, which are amplitudes needed by the electronic circuits which perform the analog-to-digital conversion and control the audio and video output displays. Gain refers to the increase in amplitude of the action potential as a result of electronic amplification. Sensitivity refers to the display resolution of the amplified signal and is expressed in microvolts or millivolts per vertical screen division.

To achieve selective amplification of the neurophysiologic signal while rejecting environmental noise, a special amplifier known as a differential or instrumentation amplifier is used (Fig. 4). A differential amplifier measures the difference in the

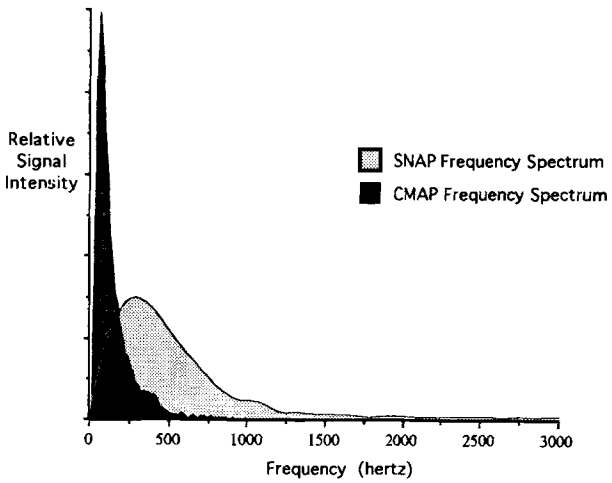


FIGURE 2. Average frequency power spectrum of the SNAP and CMAP from 9 normal subjects using filter settings of: CMAP 2–10,000 cps and SNAP 2–3000 cps.

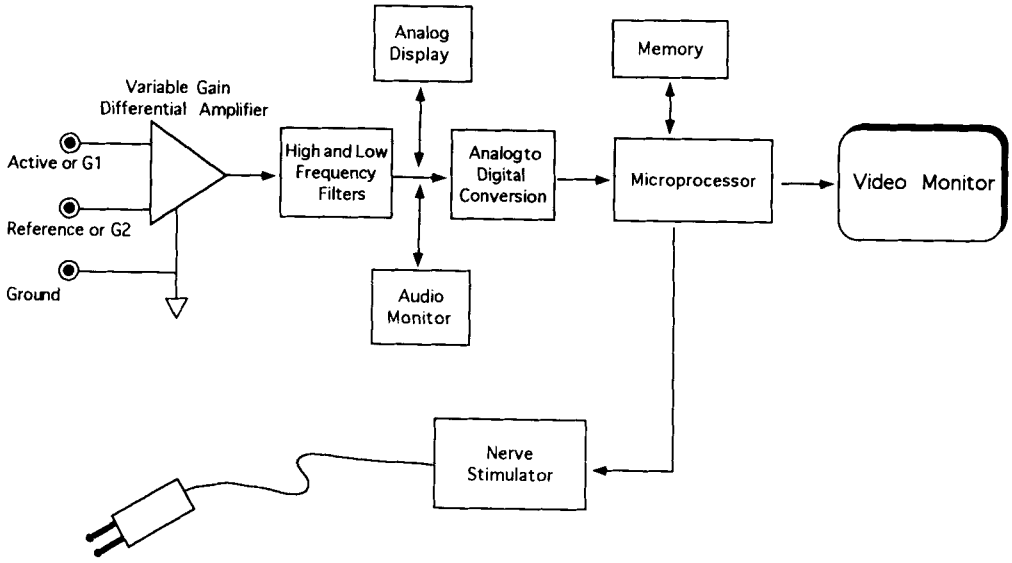


FIGURE 3. Most contemporary EMG/NCV/EP instruments use computer controlled data acquisition and display. Recording an action potential requires a series of steps consisting of amplification of the low level neurophysiologic signal, filtering out unwanted noise, conversion to digital values, scaling and analysis by the microprocessor, and finally display of the waveform on a video monitor. Using modern electronics, this process can be accomplished very rapidly (in milliseconds) allowing for "real time" display of signals.

electrical potential between the active and reference electrode inputs. If the recording electrodes are spaced reasonably close to each other and have similar characteristics, environmental electrical noise such as 60 cps power line interference will appear at both inputs with nearly the same amplitude. A signal that appears with equal amplitude at both inputs is considered "common" to the inputs and is termed a common mode signal. Since the differential amplifier only responds to the difference between the signals, most of this common but undesired signal is not amplified and is eliminated. With the same electrode setup, the active electrode is intentionally placed close to the action potential generator while the reference electrode is placed in an "electrically silent" area relative to the action potential generator. Thus at the same time that the amplifier is rejecting the common mode noise, the action potential itself will be amplified since the difference in this signal as seen by the active and reference inputs is large.

The ability to selectively amplify only the difference signal while rejecting the common mode signal is expressed by the common mode rejection ratio (CMRR). The CMRR is the ratio between the gain of the neurophysiologic potential (i.e., the difference signal) and the gain of the 60-cycle noise (i.e., the common mode signal). There are practical limits to the amount of common mode rejection which can be designed into amplifiers. Variability

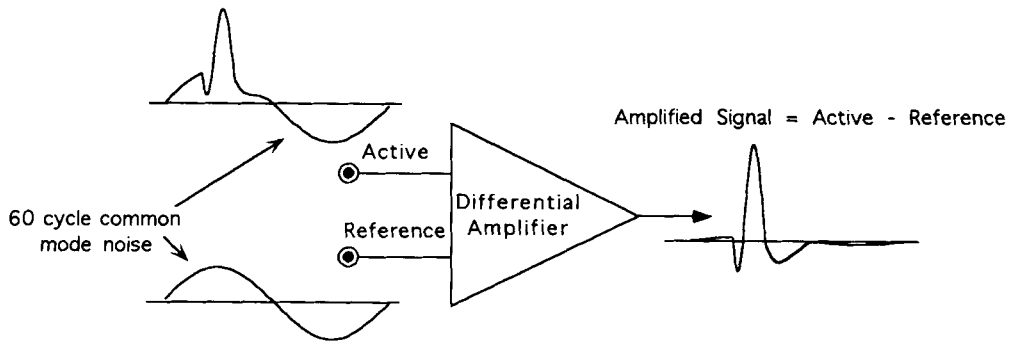


FIGURE 4. Differential amplification. The common mode 50/60-cps interference is present at both the active and reference inputs. The waveform of interest is seen by the active lead but not the reference. The differential amplifier only amplifies the difference in the signals present at the two inputs, thus eliminating most of the 60-cycle noise while retaining the waveform.

in electronic component tolerances and the presence of stray capacitance in circuits typically limit CMRR to the range of 300,000 or less.

CMRR is usually specified in units of decibels (dB) where:

$$\text{CMRR (dB)} = 20 \log \left(\frac{\text{Gain of the Difference Signal}}{\text{Gain of the Common Mode Signal}} \right) \quad (2)$$

Contemporary EMG amplifiers used for clinical studies should have a CMRR of 90 dB or greater.

During and following amplification, the neurophysiologic signal is filtered to remove extraneous and unwanted noise components. The filters will determine how the instrument responds to both rapidly changing (high-frequency) and slowly changing (low-frequency) signal components. The filtered and amplified signal is then displayed. Two general approaches are used for waveform display: analog and digital. In analog displays, the signal is displayed on a cathode ray tube as a continuously changing waveform. With digital display, the neurophysiologic potentials must be converted into a digital or numeric format before these signals can be analyzed. The conversion process is called analog-to-digital conversion (ADC) and is accomplished with special integrated circuit devices.

FREQUENCY RESPONSE AND FILTER SETTINGS

General Filter Considerations. Filtering is an integral step in the processing of neurophysiologic potentials. The most important function of filters is noise attenuation. Noise (which is always present) can be removed if the noise frequency is sufficiently different from the frequency components that make up the neurophysiologic waveform itself. Successful noise reduction improves the quality of a waveform and makes clinical interpretation easier and more accurate. A second role for filtering is to reveal information in a signal that may not be intuitive or even measurable under ordinary recording conditions. An example of this is single fiber EMG (SFEMG), in which the low-frequency filter cutoff is raised to 500 cps to remove the low frequency, slowly changing the waveform components. This decreases the pickup volume of the electrode, allowing the SFEMG needle to selectively record from the few muscle fibers immediately adjacent to the exposed needle tip to obtain jitter and density measurements.

The range of frequency components contained in a neurophysiologic potential can be defined by its frequency bandwidth, i.e., the range of frequencies from the low-frequency limit to the high-frequency limit. Filters are described by their effect on various frequency components. Low-frequency filters (LFF) remove low-frequency, slowly changing components of a waveform. Since they allow the higher frequencies to pass through and be included in the waveform they are also known as high-pass filters. High-frequency filters (HFF) remove the high-frequency, rapidly changing components and allow the low frequencies to pass through and hence are also known as low-pass filters. The cutoff frequency of a filter defines the frequency at which half of the signal energy has been removed (Fig. 5). As an example, an LFF with a cutoff of 10 cps will attenuate the amplitude of a 10 cps signal by approximately 30% (which corresponds to an energy loss of one half).

Filters are generally not perfect in that the re-

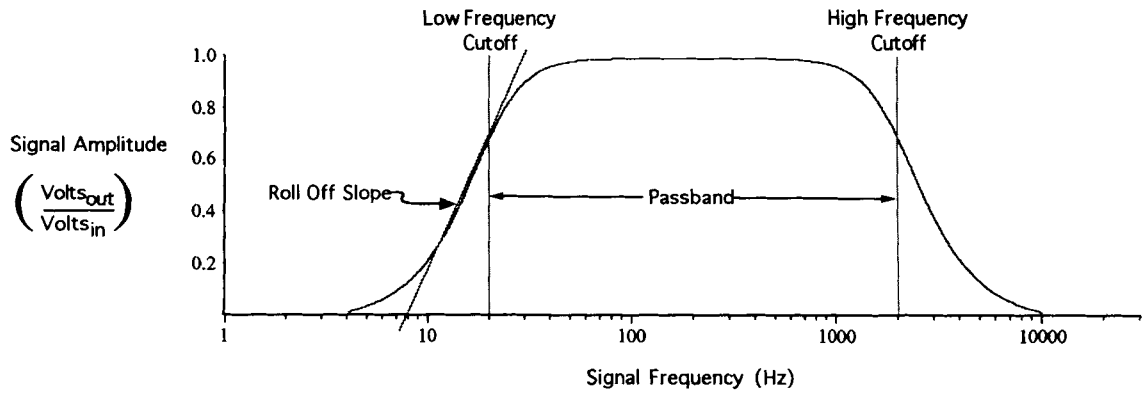


FIGURE 5. Frequency response of a 20-cps LFF in combination with a 2000-cps HFF. The region between the filters' cutoffs is known as the pass band or bandwidth. Near the upper and lower cutoffs, the attenuation occurs gradually. For the HFF, this means that some signal amplitude below the high-frequency cutoff is lost and also that considerable frequency components above the cutoff are allowed through the filter. A similar effect occurs with the LFF.

moval of unwanted frequencies occurs in a gradual fashion. The rate of rolloff of a filter reflects how rapidly it attenuates frequencies beyond its cutoff value. This rolloff rate is expressed by the order of the filter, with higher order filters having a more rapid rolloff and hence better selectivity. In addition to their effect on attenuating the amplitude of different frequencies components, a filter will change the phase of a signal passing through it. Phase shift is not particularly intuitive but can be thought of as a time delay occurring in a particular frequency component as it passes through the filter. If all the frequencies were delayed equally the effect would be unimportant, but filters can, and indeed most analog filters do, cause different amounts of phase shift in the frequency components near the cutoff value. This can result in waveform distortion. Further discussion of phase shift and rolloff are beyond the scope of this review and can be found in most general electronic texts including Barry[1] in AAEM Minimonograph #36.

Filters used in electrodiagnostic studies can be implemented as analog electronic circuits using combinations of resistors, capacitors, and amplifiers or as mathematical algorithms (digital filters) applied to the signal after conversion into a digital form.[14] If the characteristics of the filters, i.e., cutoff, rolloff rate, and phase response, are the same, then analog and digital filters will have essentially the same effect on the waveform.

All filters distort waveforms to varying degrees because of their frequency attenuation and phase shift characteristics. Because filters are used routinely, however, electromyographers become accustomed to these "distorted" waveforms and focus on detecting differences which indicate abnormality. As the use of digital signal processing in electrodiagnostic instruments continues to increase, digital filtering will become more routine. Because digital filters can be designed to have characteristics that are extremely difficult to implement in circuit form, these filters can extract additional waveform features that may not be immediately intuitive or obvious. Potential dangers also exist with digital filters if distortions created by the filtering process are misinterpreted as having a physiologic basis.[13,14]

Recommended settings for routine clinical studies are summarized in Table 1.

Table 1. Recommended filter settings for clinical studies.

Test	Low-freq setting (cps)	High-freq setting (cps)
Motor NCV	2–5	10k
Sensory NCV	5–10	2–3k
EMG—insertional activity	10–20	10k
EMG—quantitative analysis	2–5	10k
SSEP	5–20	1–2k

High-Frequency Response. The length of time during which a rapid change in potential occurs, usually from a baseline value, is the rise time of the potential. The high-frequency response of the amplifier will determine the minimum rise time for which it is able to detect the full potential change. Figure 6 demonstrates the distortion in an MUAP which occurs if the high-frequency response is inadequate to capture the rising slope of the main negative spike of the potential. As the high-frequency filter cutoff is sequentially reduced from 10,000 cps to 500 cps, there is a progressive decline in the amplitude of the waveform. Amplitude loss and elimination of rapid changes in the MUAP can also reduce polyphasia. Since rapid rise times occur when recording MUAPs with a concentric needle, an 8000–10,000-cps high-frequency limit is needed for quantitative analysis.[5]

Similarly, a 16–25% reduction in the main spike amplitude of the SNAP[5,9,19] occurs when the high-frequency response is lowered from 10,000 cps to the 500 cps. This effect is shown in Figure 7, where the inability to follow the rapidly rising and falling components of a SNAP can be easily observed. This results in a lower amplitude but also a longer latency to the peak of the waveform. Long-duration potentials, such as the CMAP and SEP,

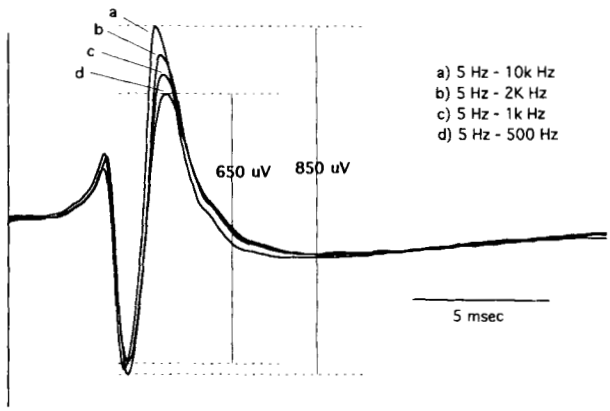


FIGURE 6. The effect of lowering the HFF cutoff on MUAP amplitude. As high-frequency components are removed by lowering the high-frequency cutoff from 10,000 cps to 500 cps, the rising slope of the main negative spike is attenuated, resulting in a 24% reduction in peak-to-peak amplitude.

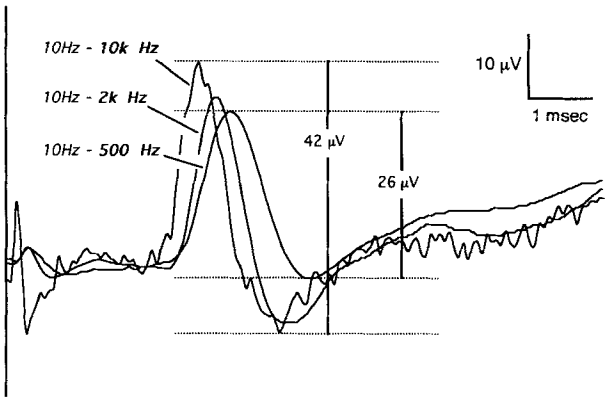


FIGURE 7. Changes in the median SNAP recorded with high-frequency filter settings ranging from 10,000 cps to 500 cps. The loss of the higher-frequency components of the waveform results in a slower risetime and a reduced amplitude. A progressive delay in the peak latency of 0.5 ms also occurs as the filter cutoff is lowered.

contain mostly low-frequency components. For these waveforms, the amplitude remains stable over a wide range of HFF settings ranging from 10,000 cps down to 500 cps.[9]

Latency to the onset of a waveform can also be affected by an inadequate high-frequency response. In this situation, the measurement system cannot follow the normal abrupt departure from baseline and the subsequent rising slope of the negative spike of the waveform.[7,19] Because of the slower rise of the waveform caused by the inadequate HFF, more time elapses before the signal reaches a voltage level that can be visually identified as the departure from baseline (Fig. 8).

Maintaining the high-frequency response at 10,000 cps would always ensure that the fast changing events in the waveform are preserved; however, significant high-frequency noise originates in the amplifier electronics and can obscure sensory evoked potentials. Decreasing the high-frequency response to 2000 cps removes most of this noise, allowing for better visualization of the SNAP (Fig. 9). The tradeoff that must be accepted for the noise reduction in sensory studies, however, is a 10–15% loss of SNAP amplitude.[5] In most clinical settings this is a reasonable compromise. If extremely low-noise amplifiers are available or averaging is used to remove noise, a higher frequency cutoff of 5000 cps can be used.

The body itself, namely the soft tissue volume conductor between the nerve or muscle fiber and the electrode, functions as a high-frequency filter. As the signal passes through body tissue, high-frequency features of the waveform, i.e., rising and falling edges and rapidly changing spikes, will

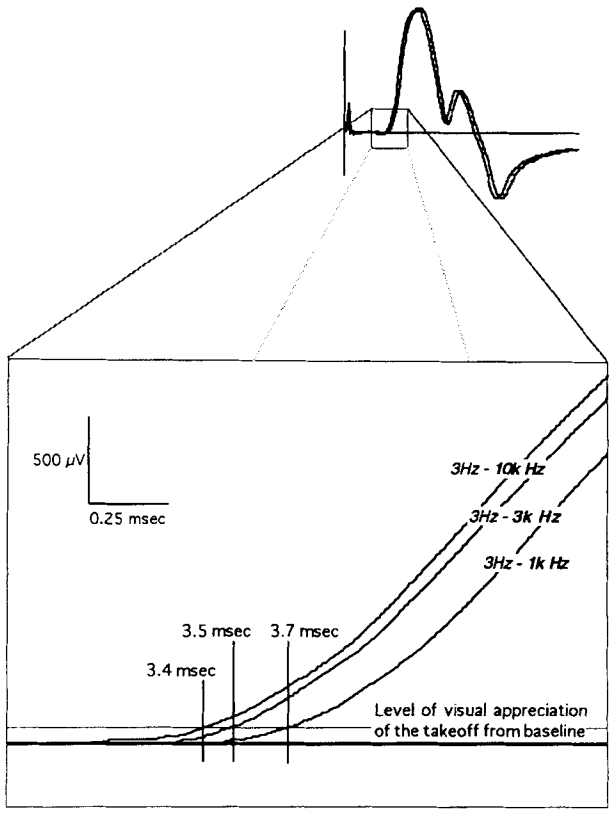


FIGURE 8. Onset of a median CMAP recorded with high-frequency filter settings ranging from 10,000 cps to 1000 cps. The loss of the high-frequency response results in a slower departure from the baseline and a less steep rise of the negative spike. The waveform takes longer to reach a voltage level that can be distinguished from the baseline, resulting in a latency delay of 0.3 ms between recordings performed at 10,000 cps and 1000 cps.

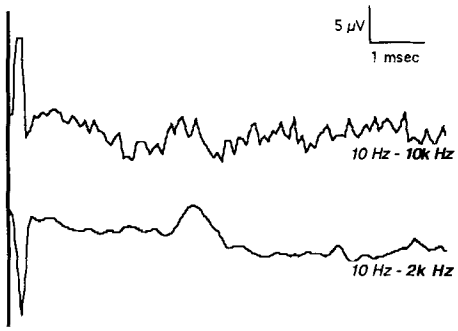


FIGURE 9. Electronic noise is generated and amplified along with any neurophysiologic potentials. The SNAP illustrated is small and requires high gain for adequate visualization. In the top tracing, using a HFF cutoff of 10,000 cps, the action potential is present but obscured by baseline noise. Decreasing the cutoff to 2000 cps removes most of the noise, allowing for easier waveform recognition and measurement.

be attenuated. A comparable way of viewing the tissue-filtering effect is to realize that low-frequency components pass through tissues with less attenuation than occurs with the high-frequency components. As a result, low-frequency action potential components can be recorded even with the electrode at a distance from the nerve or muscle cell. This effect is most obvious as one observes the blunting in the shape of the motor unit potential as an EMG needle is moved away from the motor unit.

Low-Frequency Response. The low-frequency response determines the ability to accurately record the slowly changing components of action potentials. CMAPs, evoked potentials, and the slow return to baseline of the MUAP are affected by the low-frequency response characteristics of the instrument. Inadequate low-frequency response can lead to errors in waveform amplitude, latency, duration, and morphology.

Progressively raising the low-frequency cutoff while recording a CMAP potential will cause initial amplitude loss and finally waveform distortion (Fig. 10). The recording system can no longer track the slow waveform changes near the negative peak and begins to decay back to the baseline, causing a reduction in the amplitude and duration of the main negative phase. Likewise, the slow return to baseline cannot be tracked and the waveform begins to decay sooner, distorting the terminal components of the waveform. Similar effects occur during sensory studies[9,19]; however, the LFF cutoff must be raised further for a comparable reduction in amplitude. This can be explained by the shorter duration of the SNAP which is the result of its signal content being contained in higher-frequency components than occur in the CMAP potential.

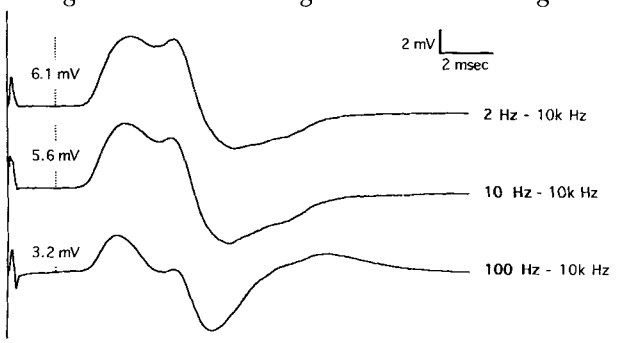


FIGURE 10. The long-duration CMAP contains predominantly low-frequency components. Raising the LFF filter cutoff from 2 cps to 10 cps initially causes a small (8%) reduction in the negative spike amplitude. With further raising of the LFF cutoff to 100 cps, the amplitude is markedly reduced, the duration of the negative spike decreases, and an extra phase is added, distorting the waveform configuration.

Raising the low-frequency cutoff also causes a decrease in the latency to the peak of a waveform (Fig. 11). The peak of the CMAP, and to a lesser extent the SNAP, are relatively slowly changing events. By removing the slowly changing components of the waveform, the instrument is unable to track the waveform peaks and begin an early return to the baseline, shifting the peak latency to a shorter value.

Motor unit potentials are also affected by changes in the low-frequency response. From a clinical standpoint, the low-frequency response is both a potential source of error and a useful way of extracting additional MUAP information. Similar to the terminal distortion seen in CMAP recordings (Fig. 10), the MUAP duration increases and an extra phase at the end of the potential can occur as the low-frequency cutoff is raised from 2 to 20 cps (Fig. 12). This type of distortion led to the recommendation by Buchthal[4] of a low-frequency limit of 2 cps for quantitative MUAP analysis.

If the LFF is further raised to exclude frequencies below 300–500 cps, there is a marked reduction in waveform amplitude and duration (Fig. 13). This effect can be used clinically to assess motor unit stability and has been termed "pseudo SFEMG." By raising the LFF cutoff, the electrical contribution from muscle fibers of the motor unit distant from the electrode is lost. This occurs because of the soft tissue filter effects previously described. Clinical applications of this technique are discussed further in the delay line and trigger section.

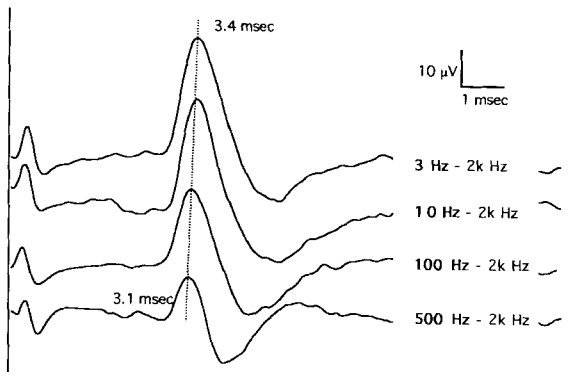


FIGURE 11. Effect of the low-frequency response on the SNAP. Amplitude and latency remain stable as the low-frequency cutoff is changed from 3 cps to 10 cps. With further increases in the LFF settings to 100 cps and 500 cps, there is a progressive shift to shorter peak latencies (0.2 and 0.3 ms, respectively) and reduced amplitudes.

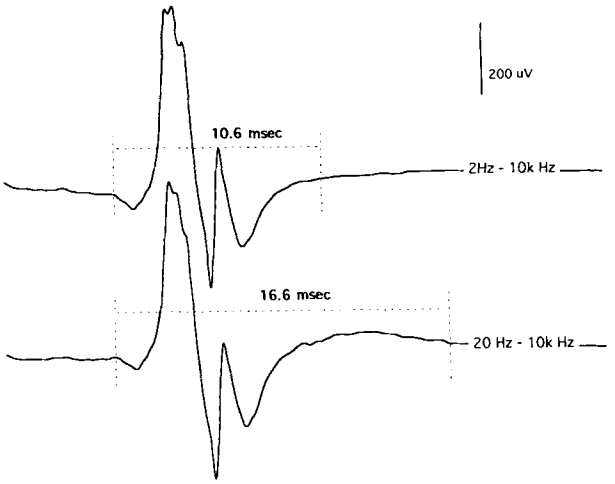


FIGURE 12. Effect of the low-frequency response on the MUAP amplitude. As the low-frequency response cutoff is raised from 2 cps to 20 cps, terminal distortion of the MUAP is seen. The addition of a trailing negative hump results from phase distortion by the filter and erroneously increases the duration of the potential from 10.6 to 16.6 ms.

Studying insertion activity with a low-frequency response of 2 cps can be difficult because movement artifacts (slowly changing polarization potentials) receive too much amplification, causing the baseline to wander. Switching the low-frequency cutoff to a higher value (10–20 cps) allows better visualization of spike potentials during insertional activity analysis. Shifting back to a lower value preserves waveform morphology when examining voluntary MUAPs.

Most potentials of clinical interest contain significant frequency components near 60 cps. Although it is tempting to use the 60 cps notch filters which are available on many instruments to decrease noise, this results in an unacceptably high degree of waveform distortion.

ELECTRODE FACTORS

In all electrodiagnostic applications, three electrodes are applied to the patient. Two electrodes, the active and reference, serve as locations between which the action potential is measured and amplified. The third, a ground electrode, serves as a zero-voltage reference point for the amplifiers.

Both surface and needle electrodes are made from a variety of metals and alloys including stainless steel, platinum, silver–silver chloride, nickel–chromium, and silver and gold alloys and plating. Whenever a metal electrode interacts with an electrolyte (i.e., electrode paste, sweat, or extracellular fluid) an electrochemical reaction occurs, resulting in a separation of negative and positive charges at the electrode–electrolyte interface.[10,22] The charge separation gives rise to an electrode polarization potential which acts like a small battery and can range in amplitude from 100 to 600 mV[21,22] depending on metal type, the electrolyte composition, and ionic concentration. With high-quality electrodes in good condition, polarization potentials are relatively stable and do not appear since the amplifier is not responsive to D.C. (unchanging) signals. Following movement of a needle or surface electrode, the polarization potential will dissipate and fluctuate for a short period of time, giving rise to spike potentials and baseline movement artifacts. When electrodes are dirty, corroded, or in the case of monopolar needle electrodes, the Teflon coating is frayed or excessively thinned, polarization potentials are unstable. Unstable polarization potentials can sometimes give rise to artifacts which resemble abnormal membrane potentials. Silver–silver chloride electrodes earned their reputation because their electrode polarization potentials are quite stable and hence relatively noise-free.

The electrode surface has an electrical capacitance and a resistance component. Both contribute to the total impedance of the electrode.[20] This impedance is frequency-dependent and increases as electrode area decreases. The importance of this variation of impedance with frequency and the interaction with the amplifier will be discussed later.

Surface Electrodes. Most NCV and SEP techniques utilize small surface disk electrodes or ring electrodes for the digits as the two electrodes between which the action potential is measured. In motor nerve conduction studies, the active electrode is placed over the motor point (generally midway between the origin and insertion of the

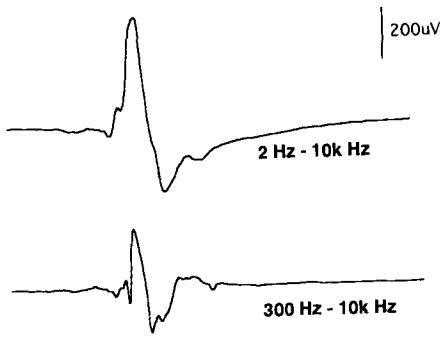


FIGURE 13. "Pseudo" SFEMG using a standard monopolar needle. Raising the low-frequency filter to 300–500 cps decreases the pickup area of the electrode, causing a marked reduction in MUAP amplitude and duration, and alters waveform shape.

muscle) and the reference electrode over the distal tendon or perhaps even further away. By placing the electrode over the motor point (motor endplate region) of the muscle, depolarization and action potential (AP) generation will occur directly beneath the active electrode. The volume conduction of this AP to the electrodes will give rise to a biphasic potential with an initial negative deflection[8] and a relatively easily determined takeoff. In contrast, when the active electrode is moved away from the motor endplate region, a small positive deflection precedes the main negative waveform spike. Accurate assessment of onset is more problematic in this situation but remains best measured to the initial departure from baseline.

The location of the reference electrode will also influence the shape and amplitude of the CMAP. Ideally, the reference is placed "far enough away" so that its potential will remain close to zero. Because of the large size of muscle compound action potentials, this is frequently not the case. When both the active and reference electrodes "see" the potential at the same time, the recorded waveform is the difference in voltage between the electrodes and has effects on amplitude and morphology (Fig. 14).[12]

In recording SNAPs, it is customary to place both electrodes over the nerve. The spacing between the active and reference electrode will affect the SNAP amplitude. As the interelectrode spacing is reduced to less than approximately 4 cm, the amplitude of the SNAP progressively declines. The 4-cm spacing tends to maximize the peak-to-peak amplitude because at this distance the traveling wave of depolarization in the sensory nerve will have completely passed the active electrode before arriving at the reference electrode. This assumes a conduction velocity in a sensory nerve of 50 m/s (50 mm/ms) and an action potential duration at the nerve membrane surface of 0.8 ms giving a length for the depolarized region of 4 cm. Hence, to use normal reference amplitude values, standardization of the distance between the two surface electrodes is necessary. The volume conduction principles which underlie this effect are discussed more fully by Dumitru.[8] Techniques for needle electrode sampling of nerve action potentials are well described by Oh.[17]

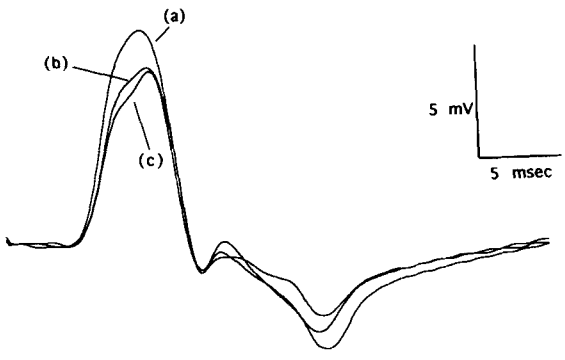


FIGURE 14. Variability in the median CMAP recorded with reference electrode placement: (a) at the metacarpal phalangeal joint; (b) at the midproximal phalanx; and (c) at the middistal phalanx. Placements (b) and (c) yield nearly identical responses while (a) has a 24% larger baseline-to-peak amplitude.

Needle Electrodes and Techniques. *Monopolar Needle Electrodes.* Monopolar EMG needle electrodes consist of a Teflon-coated stainless steel wire that tapers to a sharp tip (Fig. 15). When using a monopolar needle, the MUAP potential is measured as the voltage difference between the active needle tip which is placed into the muscle belly and a reference skin-surface electrode. In an effort to maintain a zero potential, the surface (reference) electrode is placed over a nearby tendinous or bony site. Even at this location, however, the surface electrode is seldom completely electrically silent. Depending upon the degree of muscle contraction originating from the muscle being studied or other muscles being tensed by the patient, background potentials will be recorded from the surface electrode and displayed along with those originating near the needle tip. Placing the surface reference electrode over the skin close to the needle can be helpful in reducing the influence of distant muscle contraction when searching for spontaneous activity.

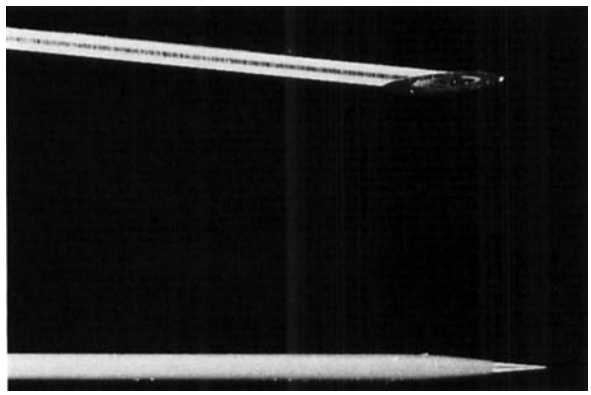


FIGURE 15. Comparison of a standard monopolar EMG needle (bottom) and concentric needle (top). The active recording surface consists of the exposed tip of the monopolar needle as compared to the central beveled core of the concentric needle.

The average exposed tip area is approximately 0.14–0.20 mm^2. With repeated sterilization, the exposed area can increase up to 20% even without obvious Teflon fraying. Since a monopolar needle records the average potential over its exposed tip, cracked, chipped, or frayed Teflon coating will cause a reduction in the amplitude of motor unit potentials. If breaks or cracks in the Teflon coating along the shaft of the electrode exist, the signal received at the tip will be "short-circuited" back into the tissue, further reducing the amplitude. Since the Teflon does not extend fully to the hub in many monopolar needles, the needle should not be inserted its entire length.

The reference electrode in the monopolar technique has approximately a tenfold lower impedance than the needle electrode. This impedance difference interferes with the amplifier's ability to reject common mode signals, often resulting in 50/60 cps interference.

Concentric Needle Electrode. The concentric needle consists of a fine wire electrode, insulated and housed inside an outer cannula or hypodermic needle (Fig. 15). Action potentials are measured as the difference between the tip of the central isolated core or active electrode and the bare metal cannula, which serves as the reference electrode. Because of the physical proximity of the active and reference recording surfaces, the reference cannula will be near the MUAP and will record an attenuated version of it.

The exposed area of the central core of a concentric electrode is determined by the wire diameter and the angle of the bevel. The active electrode tip area varies between 0.2 and 0.9 mm^2. The recording surface is usually smaller but more constant in size than the exposed tip of the monopolar needle. Because it is smaller, the core electrode impedance can be up to ten times larger than the monopolar electrode. As in the monopolar technique, the core and the cannula impedance change with frequency, decreasing at the higher frequencies. Any difference in impedance between the central core and the cannula will reduce common mode signal rejection. Because they are close together, the common mode signal impressed upon the two electrodes in the concentric application is more likely to be the same as compared to the monopolar needle/surface disk electrode pair. For this reason, common mode noise origination from power lines tends to be somewhat less problematic with concentric needle recordings.

The Influence of Needle Characteristics on MUAP Parameters. The type, size, orientation, and configuration of the needle electrode will influence the characteristics of the MUAP recorded. The recording surfaces of both monopolar and concentric needles are large relative to the diameter of a typical muscle fiber (60 μm).[2] As a result the active recording surface is in contact with many muscle fibers, some of which belong to a depolarizing motor unit, while others are electrically silent. The resulting MUAP recorded represents a summation and average of the electrical activity occurring along the entire exposed needle tip.

Clinical studies[4,6,7,11,15,18] comparing the differences in MUAP parameters between monopolar and concentric needles have shown variable results but in general have indicated that monopolar needles produce larger amplitudes and greater phasicity but comparable durations (Table 2). The lack of standardization in recording techniques and measurement criteria makes comparisons of absolute amplitude, duration, and phasicity between these studies difficult.

The reported MUAP amplitude increase of 0–100% when using monopolar needles as compared to concentric needles results from several factors. The main MUAP spike amplitude is determined by the small number of muscle fibers immediately adjacent to the electrode tip.[3,15,16] Since the active recording surface of the concentric needle is shielded on one side by the reference cannula, the active electrode does not record action potentials from muscle fibers behind the cannula.

Table 2. Value of MUAP parameters recorded with monopolar needles compared to concentric needles.

Study	#MUPs evaluated	Amplitude	Duration	Phases/turns*
Buchthal, Guld, and Rosenfalck[4]		↑	ns	
Nandedkar and Sanders[15]	200	↑	ns	↑ †
Howard, McGill, and Dorfman[11]	7270	↑	ns	↑
Chu, Chan, and Bruyninckx[7]	52	↑	↑	↑
Pease and Bowyer[18]	120	ns	ns	
Chan and Hsu[6]	202	↑	↑	ns

*Number of turns.

In contrast, the multidirectional recording nature of the monopolar needle likely places more muscle fibers from the MUAP under study close to the active electrode tip, resulting in an increase in the MUAP amplitude. A second factor relates to the previously discussed concept of differential amplification. When using concentric needles, the reference cannula will be in the physical vicinity of the actively depolarizing motor unit and will record an attenuated version of the MUAP. Since the recorded potential is the difference between the active tip and the reference cannula, the attenuated version of the MUAP will subtract from the MUAP seen by the active tip and result in a decrease in the amplitude of the potential.

Motor unit complexity and phasicity are related to the degree of temporal synchrony of the discharges of the individual muscle fibers contributing to the MUAP. Variability in muscle fiber conduction velocity, endplate dispersion, and conduction time in terminal axon branches all contribute to this synchrony. Monopolar needles record activity from more muscle fibers than occurs with concentric needles because of their generally larger exposed tip and the distant location of the reference electrode. This larger population of discharging muscle fibers results in less temporal synchrony and a corresponding greater degree of polyphasicity.

Lastly, MUAP duration reflects the total number of muscle fibers within the motor unit or at least within a recording volume that is large relative to the region responsible for determining motor unit amplitude. Duration measurements are greatly influenced by the slowly changing leading and terminal portions of the potential. Because body tissue acts as a high-frequency filter, the amplitude of rapidly changing waveform components diminishes rapidly while slowly changing components pass relatively unattenuated. This allows many muscle fibers to contribute to the slowly changing leading and terminal portions of the MUAP. As a result, the size and orientation of the needle tip play a less important role in duration measurements than for other parameters. Although the majority of studies have not demonstrated any significant differences in MUAP duration between needle types, two reports have found monopolar durations to be longer. Chu et al.[7] found monopolar durations to be statistically prolonged though the magnitude of the increased duration was only 10–20%. Chan and Hsu,[6] using a technique in which simultaneous recordings of the same MUAP were made with a concentric and monopolar needle, found significantly longer durations with the monopolar recording. This was attributed to the larger area of the monopolar needle and the distant reference location which limits the effect differential amplification has on canceling muscle fiber potentials discharging at a distance from the needle electrode.

SUMMARY

This portion of the monograph has explored action potential generation, the frequency content of waveforms, basic principles of signal amplification, distortions in waveforms that can result from inadequate high- and low-frequency filter settings, and the influence of electrode characteristics on waveform amplitude and morphology. An appreciation of the sources and magnitudes of the errors due to technical recording limitations, and importantly, techniques which can be used to minimize their effects, will improve the electrodiagnostic physician's ability to acquire accurate data.

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